

Introduction To Manufacturing Technology (4FTC2030)

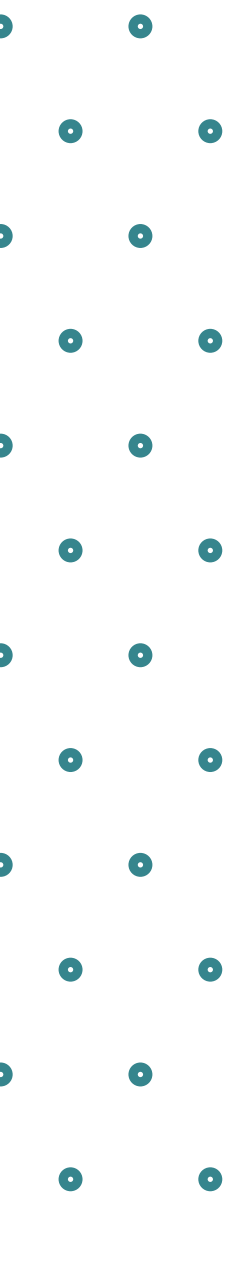
Project : Machining & CAM Technical Report

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1.Introduction

This section presents the design and fabrication of a hydraulic-powered robotic arm developed to demonstrate the principles of fluid power and mechanical motion. The system operates using hydraulic actuation through syringes and tubing to produce controlled movement without the use of electrical components.

The robotic arm is designed to perform basic pick-and-place operations by gripping, lifting, and transferring lightweight objects. The project highlights the application of manufacturing techniques, material selection, and assembly processes in developing a functional mechanical system.

2.Objective

- The main objectives of this project are:
- To design and construct a hydraulic-powered robotic arm using lightweight materials.
 - To demonstrate controlled motion using hydraulic actuation without electrical assistance.
 - To develop a system capable of gripping, lifting, and transferring lightweight objects.
 - To apply basic manufacturing and assembly techniques in building the system.
 - To evaluate the performance of the robotic arm in terms of movement, stability, and control.

3.Project Management

3.1 Work Breakdown Structure

The project was divided into a series of structured tasks to ensure efficient planning and execution. The main stages of the project are summarized below:

Task no.	Task Description
1	Design of robotic arm
2	Material selection
3	Fabrication
4	Assembly
5	Testing and evaluation

Table 1.1.0

3.2 Gantt Chart

A simple timeline was developed to organize the project activities and ensure completion within the allocated time.

Task	Day 1	Day 2	Day 3
Design	✓		
Material Selection	✓		
Fabrication		✓	
Assembly		✓	
Testing			✓

Table 1.1.1

4.Design Overview

The hydraulic robotic arm was designed to perform controlled pick-and-place operations using fluid power. The system consists of a base platform, multiple arm segments, joints, a gripper, and a hydraulic actuation system. The movement of the arm is achieved through syringes connected by flexible tubing. When pressure is applied to one syringe, fluid is transferred to another, causing linear motion that is converted into rotational or lifting movement at the joints.The design allows for basic degrees of freedom, enabling the arm to move vertically and horizontally, as well as to grip and release objects. The structure was kept lightweight to ensure stability and efficient operation.

The main components of the system include:

- Base platform
- Arm segments
- Joints and connectors
- Gripper mechanism
- Syringes and tubing (hydraulic system)

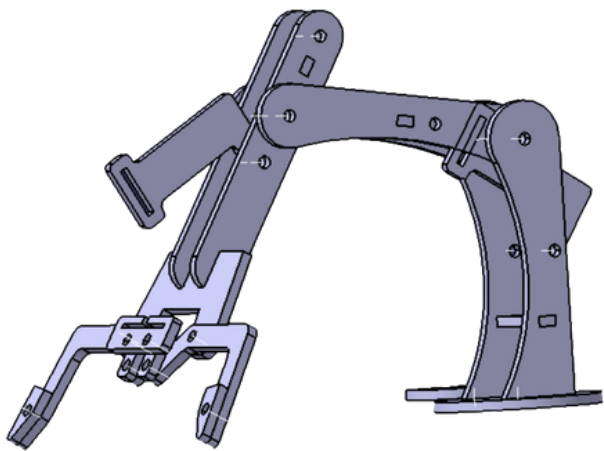


Fig 1.1.0

Figure 1.1.0: Hydraulic robotic arm showing main structural components

5. Bill of Materials (BoM)

The materials used in the construction of the hydraulic robotic arm were selected based on availability, lightweight properties, and ease of fabrication. The main components are listed in Table 1.1.2.

Component	Material	Quantity	Purpose
Syringes	Plastic	8	Hydraulic actuation
Tubing	Plastic	4	Fluid transfer
Arm segments	PLA	2	Structure of the arm
Base platform	PLA	1	Support and stability
Connectors / Joints	Steel	17	Movement between segments
Adhesive (Glue)	–	–	Assembly of components
Zip tie	Plastic	10	Parts handling

Table 1.1.2

6.Manufacturing and Assembly

The hydraulic robotic arm was fabricated using a combination of simple manufacturing techniques and manual assembly. Lightweight materials were shaped and assembled to form the structure, while hydraulic components were integrated to enable controlled motion.

6.1 Fabrication Process

- The structural components of the robotic arm were produced using lightweight materials such as PLA and cardboard. Cutting and shaping were carried out manually and, where applicable, using 3D printing techniques to achieve accurate dimensions.
- Care was taken to ensure that all parts were properly aligned and sized to allow smooth movement at the joints.

6.2 Assembly Process

The assembly process was carried out in a step-by-step manner as follows:

- The base platform was constructed to provide stability.
- Arm segments were connected using joints to allow movement.
- The gripper was attached at the end of the arm.
- Syringes were fixed at different joints and connected using tubing.
- The hydraulic system was filled with fluid and tested for proper operation
- The whole Hydraulic system was then connected to a joystick for controlling the movement.



Fig 1.1.1

Fig 1.1.1: Hydraulic control mechanism (joystick) used to regulate fluid flow and control the movement of the robotic arm

6.3 Techniques and Precautions

several techniques and precautions were followed to ensure proper functionality and safety:

Techniques:

- Proper alignment of joints to allow smooth motion
- Secure fixing of syringes to prevent displacement using zip tie.
- Ensuring tight connections in tubing to avoid leakage

Precautions:

- Careful handling of cutting tools
- Avoiding excessive force during assembly
- Ensuring no air bubbles in the hydraulic system
- Checking for leaks before operation



Fig 1.1.2

Fig 1.1.2 : Shows use of cable ties (zip ties) for securing components and maintaining stability of the hydraulic arm structure

7.Testing and Results

The hydraulic robotic arm was tested to evaluate its performance in terms of movement, control, and functionality. The system was able to successfully perform basic pick-and-place operations by gripping and transferring lightweight objects.

The movement of the arm was smooth and responsive to the applied hydraulic pressure, demonstrating effective transmission of force through the fluid system. The gripper was capable of holding objects securely, and the overall structure remained stable during operation.

However, minor limitations were observed, including slight delays in response and occasional leakage in the hydraulic connections. These issues did not significantly affect the overall performance but indicate areas for potential improvement.



Fig 1.1.3

Fig 1.1.3 Fully assembled hydraulic robotic arm demonstrating controlled motion and object manipulation

6. Conclusion

This project successfully demonstrated the integration of mechanical design, manufacturing processes, and computer-aided engineering through the development of a hydraulic robotic arm and the machining of a precision axle component.

In Part I, the hydraulic robotic arm was designed and assembled using lightweight materials and hydraulic actuation principles. The system was able to perform controlled pick-and-place operations, demonstrating the effectiveness of fluid power in generating motion without the use of electrical components.

In Part II, the axle component was manufactured through a sequence of machining operations including cutting, turning, drilling, and threading. While the overall process was successfully completed, measurement results showed that the length was accurate, whereas the diameter exhibited a deviation of approximately 3–4 mm, highlighting the importance of precise control during turning operations.

The use of CAM software enabled accurate simulation of the machining process, optimization of toolpaths, and estimation of machining time, significantly improving process planning and reducing potential errors.

Overall, the project enhanced practical understanding of both traditional manufacturing techniques and modern digital tools. It also emphasized the importance of accuracy, proper setup, and process control in achieving high-quality engineering outcomes. Future improvements may include the use of CNC machining to improve dimensional accuracy and consistency.

Appendix

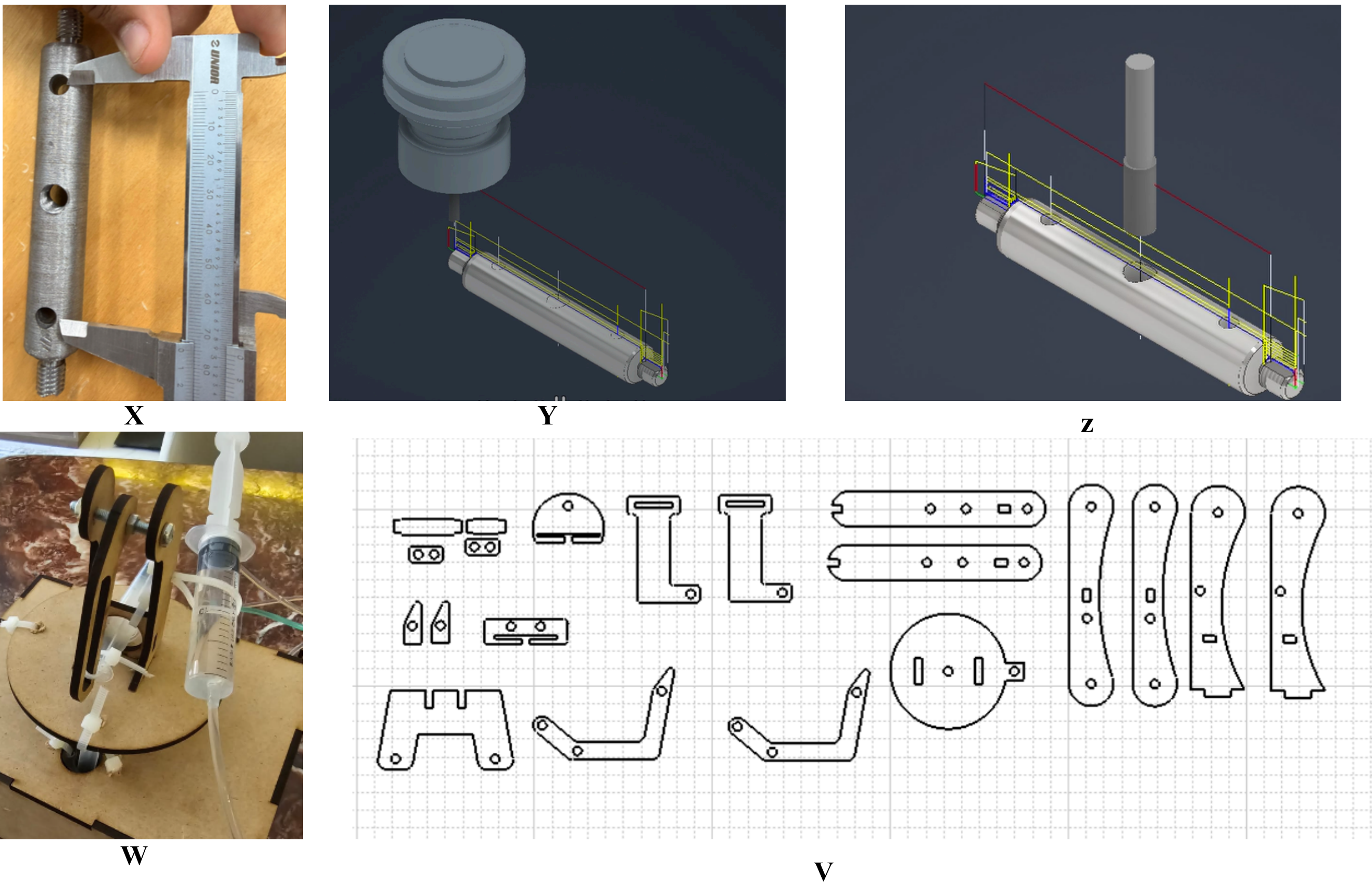


Figure X: Measurement of component length using a vernier caliper
Figure Y: CAM simulation showing tool setup and machining environment
Figure Z: CAM simulation of drilling operation and toolpath execution
Figure W: Hydraulic robotic arm assembly showing syringe-based actuation system
Figure V: CAD design layout of hydraulic arm components prior to fabrication